

Machines, Not Dials:

Chronological Unit Commitment and a Reconstructed Spatial Demand Field for the O‘ahu Power System

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Abstract

A companion study (Set 8) found that the distributional assumption placed on nodal demand relocates, rather than merely inflates, estimated adequacy risk in a DC optimal power flow model of the O‘ahu grid. That study represented generators by a single box constraint, $0 \leq P_g \leq P_g^{\max}$, and sampled hours independently with replacement. Both choices are load-bearing: the first removes every operating limit a thermal machine actually has, and the second destroys the temporal adjacency that such limits act upon. This paper rebuilds the experiment as a chronological mixed-integer unit commitment problem over 8,760 consecutive hours, adding minimum stable generation, ramp limits, minimum up and down times, and start-up costs one constraint at a time.

Two results follow. First, the binding constraint on this system is *ramp*, not capacity: at a reserve margin of 43% the fleet sheds energy in 0.61% of hours once hour-to-hour movement is restricted, and in no hours before that. The shortfall is one of speed, and the earlier model was structurally blind to it. Second, and more consequentially for method, we show that the spatial demand field used by Set 8 — and by most studies that have only a system-level load series — is *rank one*: a fixed share vector scaled by a scalar time series. Its direction cannot move, yet direction is the quantity the research question is about. We reconstruct a rank-two field from three public datasets, preserving the filed system total to 3×10^{-4} MW while allowing the share of load at the urban node to swing from 52.7% to 84.1% across the year, a diurnal rotation that emerges without being imposed.

We report two errors found by self-audit during this work, with their quantified effect, and a 600-run sensitivity protocol that converts the principal remaining judgement call into a measured statement. We conclude that every *directional* result here is robust, and that no network *magnitude* should be read as a claim about the physical O‘ahu system, because the network representation is entirely assumed.

1 Introduction

Island power systems are the cleanest available laboratory for adequacy questions. O‘ahu has no interconnection: if the island cannot generate the energy, no neighbour can supply it. Every megawatt-hour served is produced within a 1,508 MW fleet of 24 thermal units at six sites.

Set 8 of this project asked what happens to computed adequacy risk when the analyst’s distributional assumption about demand is wrong. The answer was that misspecification does not merely scale the risk estimate; it moves the risk to different nodes. That finding rested on a dispatch model in which a generator was a single interval constraint. A 128 MW steam unit could idle at 3 MW, reach full output within the hour, and cycle on and off at no cost.

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This is not a small idealisation, and it is not obviously conservative. It is not clear *a priori* whether removing operating limits makes a system look safer (any unit can always be called upon) or more dangerous (no unit is ever forced to stay on). The present paper resolves that question empirically and finds something more interesting than either: the limits create an entirely new failure mode, invisible to the earlier model, that has nothing to do with installed capacity.

Contributions.

1. A chronological MILP unit commitment formulation over the full year (Section 3), with an explicit treatment of the start-up ramp that is *necessary* rather than refining — omitting it renders every steam unit on this fleet permanently unstartable (Section 3.2).
2. A reproducible reconstruction of a spatially and temporally resolved demand field from public data, raising the demand matrix from rank one to rank two while preserving the filed system total exactly (Section 4).
3. A constraint ladder that attributes every movement in the results to a single added restriction (Sections 5 and 6).
4. A self-audit reporting two errors in our own construction with their quantified effect, and a sensitivity protocol over 600 perturbed model variants (Sections 7 and 8).

A note on epistemic labelling. Throughout, quantities are marked **FILED** when they originate in a government form or a regulatory filing, and **ASSUMED** when they are our own judgement. This is not decoration. Section 9 shows that the two findings of this paper rest on layers of very different evidential quality, and the distinction determines which of them may be quoted as a statement about O‘ahu.

2 The O‘ahu test system

2.1 Generation

The fleet is taken from EIA Form 860 (2024, summer capacity) restricted to utility and contracted firm O‘ahu plants: Kahe, Waiau, H-POWER, Kalaeloa, Campbell Industrial Park, and Schofield Generating Station. Airport emergency generators and refinery self-generation are excluded, as are solar, wind and battery resources, whose effect is already embedded in the net load series. Kalaeloa’s two combustion turbines and steam tail are aggregated as a single 220 MW combined-cycle resource; dispatching the steam tail independently would be physically misleading.

Table 1: Composition of the modelled fleet. Capacities are **FILED**; operating limits are **ASSUMED** by technology class.

Technology	Units	Capacity (MW)	$P^{\min}/P^{\max}$	Ramp (%/h)
Oil steam	12	936.3	0.40	30
Combined cycle	1	220.0	0.40	50
Combustion turbine	3	215.4	0.00	100
Municipal waste	2	86.0	0.80	20
Internal combustion	6	50.4	0.00	100
Total	24	1,508.1		

Half the fleet by unit count, and 62.1% of it by capacity, is oil steam — the class combining a high minimum stable generation with the slowest ramp. The genuinely flexible plant (combustion turbines and reciprocating engines, which may start from cold and move at full rate) totals only 265.8 MW, or 17.6% of capacity. This structural fact drives the principal result of Section 6.

2.2 Demand

The load series is the 2021 base scenario of Hawaiian Electric’s Integrated Grid Planning O’ahu Inputs Workbook 3, filed 2022-03-31 in PUC docket 2018-0165. It is assembled from ten hourly layers as

$$\text{gross} = \text{underlying} + \text{managed EV} - \text{base forecast} + \text{future eBus},$$

with net load subtracting distributed PV, four storage sectors, efficiency and time-of-use shifting. The result is 8,760 observations with mean 743.9 MW, minimum 472.6 MW and peak 1,054.3 MW.

Remark 2.1 (Filed is not metered). This series is a *filed planning profile*, not a meter reading. It was submitted under oath to the regulator, so it is a real document; but it records what the utility modelled, not what the system recorded. We use “filed” rather than “measured” throughout, and recommend the same discipline to anyone quoting these results.

The reserve margin implied by Table 1 against the annual peak is

$$\frac{1508.1 - 1054.3}{1054.3} = 43.0\%.$$

This number should be held in mind throughout Section 6: the system is not short of installed capacity.

2.3 Network

The network is a stylised nine-bus representation: the seven O’ahu judicial districts as published by the Census as county subdivisions, with the Ewa district split into EWA-WEST, EWA-CENTRAL and KAHE. The split is necessary: point-in-polygon assignment on EIA-860 coordinates places all six modelled plants inside Ewa, which spans roughly twenty miles, so leaving it whole would collapse all island generation onto one node and erase precisely the injection geography the study exists to examine. Ten branches are wired as the two-corridor ring (one north, one south) that Hawaiian Electric describes in public material, leaving two independent loops — a requirement, since a radial network makes DC-OPF and the transportation model identical.

Assumption 2.2 (The network is ours). Bus *locations* are **FILED**: population-weighted 2020 census-tract centroids, and EIA plant coordinates. Every *electrical* parameter is **ASSUMED**: the topology, line lengths (great-circle separation $\times 1.25$), reactance ($0.80 \Omega/\text{mi}$ at 138 kV on a 100 MV A base), thermal limits (200 MW per circuit) and circuit counts. Not one comes from a filing.

Assumption 2.2 is the reason Section 6.3 declines to interpret the magnitude of any congestion statistic.

3 Chronological unit commitment

3.1 Formulation

Let $\mathcal{G}$ index units, $\mathcal{N}$ buses, $\mathcal{L}$ branches and $\mathcal{T}$ the hours of a block. Binary $u_{g,t}$ denotes commitment, with start-up $v_{g,t}$ and shut-down $w_{g,t}$. Continuous $P_{g,t}$ is output, $\theta_{n,t}$ the voltage

angle and $s_{n,t} \geq 0$ load shed. The objective is

$$\min \sum_{t \in \mathcal{T}} \left[\sum_{g \in \mathcal{G}} (c_g P_{g,t} + C_g^{\text{su}} v_{g,t}) + \text{VOLL} \sum_{n \in \mathcal{N}} s_{n,t} + \epsilon \sum_g g P_{g,t} \right], \quad (1)$$

with $\text{VOLL} = 10,000 \text{ USD/MWh}$ and a tie-break term $\epsilon = 10^{-6}$ that makes dispatch unique between cost-identical units. The tie-break is a correctness device, not a cosmetic one: without it, equal-cost units receive an arbitrary split and results are not reproducible across solver versions.

Subject to nodal balance, DC flow $f_{\ell,t} = b_{\ell}(\theta_{n,t} - \theta_{m,t})$ with $|f_{\ell,t}| \leq \bar{f}_{\ell}$, an angular reference at the urban bus, and the operating constraints

$$P_g^{\min} u_{g,t} \leq P_{g,t} \leq P_g^{\max} u_{g,t}, \quad (2)$$

$$u_{g,t} - u_{g,t-1} = v_{g,t} - w_{g,t}, \quad (3)$$

$$\sum_{\tau=t-T_g^{\text{up}}+1}^t v_{g,\tau} \leq u_{g,t}, \quad \sum_{\tau=t-T_g^{\text{dn}}+1}^t w_{g,\tau} \leq 1 - u_{g,t}. \quad (4)$$

3.2 The start-up ramp is necessary, not refining

The naive ramp constraint $|P_{g,t} - P_{g,t-1}| \leq R_g$ is wrong here, and catastrophically so. On this fleet every steam unit has $P_g^{\min} = 0.40 P_g^{\max}$ but $R_g = 0.30 P_g^{\max}$. Starting up requires moving from 0 to $P_g^{\min}$ within one hour by (2), a step of $0.40 P_g^{\max} > R_g$. The unit is therefore *permanently unstartable*.

Proposition 3.1. *If $P_g^{\min} > R_g$ and ramping is enforced without a start-up allowance, then any unit off at $t = 0$ satisfies $u_{g,t} = 0$ for all t .*

Our first implementation did exactly this and reported load shedding in 95.8% of hours — an apparent collapse of the O‘ahu system that was entirely an artefact of formulation. The correct form separates the start-up movement, with $S_g = \max(P_g^{\min}, R_g)$:

$$P_{g,t} - P_{g,t-1} \leq R_g u_{g,t-1} + S_g v_{g,t}, \quad (5)$$

$$P_{g,t-1} - P_{g,t} \leq R_g u_{g,t} + S_g w_{g,t}. \quad (6)$$

Remark 3.2 (The diagnostic was economic, not numerical). The solver reported optimality at every block. No exception was raised. The error surfaced because the *dispatch pattern* was economically absurd: cheap steam capacity idle while expensive diesel engines ran continuously, with 43% of capacity unused during a purported shortage. We record this because it generalises — in mixed-integer dispatch models, implausible economics is a more sensitive detector of specification error than any solver diagnostic.

3.3 Rolling horizon

Ramp constraints couple t to $t - 1$; minimum up/down times couple a commitment decision to several subsequent hours. Set 8 sampled hours independently with replacement, so hour 5,000 could follow hour 12 and no “previous hour” existed for (5) to reference. Set 9 therefore solves the year as 365 consecutive 24-hour blocks, carrying $(u, P, T^{\text{up}}, T^{\text{dn}})$ across each boundary. This is a different experimental design, not a refinement, which is why Set 9 reuses none of Set 8’s numerical results. Blocks are capped at a 1% optimality gap and 20 seconds; capped blocks are counted and reported.

4 Reconstructing the spatial demand field

4.1 The rank-one problem

Let $D \in \mathbb{R}^{9 \times 8760}$ be the bus-by-hour demand matrix. Set 8, like essentially every study with only a system-level load series, constructed

$$D = \alpha \ell^\top, \quad \alpha \in \Delta^8, \quad \ell \in \mathbb{R}^{8760}, \quad (7)$$

with α the 2020 resident-population share vector. Then $\text{rank}(D) = 1$: the 78,840 entries carry only 8,760 degrees of freedom, the nine bus loads can only scale together, and the demand *direction* $D_{:,t}/\|D_{:,t}\|_1$ is constant in t .

This is a serious problem for the present research question. Set 8's own finding is that direction governs adequacy. Equation (7) holds direction fixed by construction, so the quantity of interest was the one quantity the data could not express.

4.2 Static sector reallocation

The first repair replaces population with a sector-weighted key:

$$\alpha_i = f_{\text{res}} \frac{\text{pop}_i}{\sum_j \text{pop}_j} + f_{\text{com}} \frac{\text{com}_i}{\sum_j \text{com}_j} + f_{\text{ind}} \frac{\text{ind}_i}{\sum_j \text{ind}_j}, \quad (8)$$

with sector energy shares f from EIA Form 861 restricted to Hawaiian Electric Co Inc, the O'ahu operating entity ($f_{\text{res}}, f_{\text{com}}, f_{\text{ind}} = 0.2675, 0.3112, 0.4213$), residential sited by Census population and the two employment sectors by LEHD LODES workplace jobs. Normalising each key separately is what makes the terms commensurable; the shares then sum to unity automatically, since each key sums to one and $\sum_s f_s = 1$.

O'ahu is a commuter island: the Honolulu district holds 39.9% of residents but 71.0% of jobs. Allocating by residence therefore exports a large block of commercial load out of the urban core in every hour. Under (8) the Honolulu share rises from 39.87% to 56.89%. The method is not simply urban-biased: EWA-WEST *gains* share (2.96% $\rightarrow$ 4.73%) despite few residents, because Campbell Industrial Park places 7.76% of the island's industrial employment there against 2.96% of its population.

But (8) is still of the form (7). The rank is still one.

4.3 Sector-resolved hourly allocation

The rank barrier is structural: any single share vector, however well estimated, yields a rank-one field. Rank rises only if sectors with *different spatial footprints* also have *different temporal profiles*. Writing $a_{i,s}$ for the share of sector s at bus i and $\gamma_s(t)$ for its system-level hourly load,

$$D_{i,t} = \sum_s a_{i,s} \gamma_s(t) - \pi_i \text{dg}(t), \quad (9)$$

where $\text{dg}(t)$ is the behind-the-meter layer and π its (assumed) spatial key. Sector profiles γ_s are built from OpenEI TMY3 reference buildings at Honolulu weather, folded into $12 \times 2 \times 24 = 576$ month/weekend/hour cells so that repeatable rhythm is retained and single-year weather noise is averaged out. The commercial profile is an energy-weighted composite of fifteen reference building types.

Two implementation details matter. Behind-the-meter PV can exceed a suburban bus's gross draw at midday, which (9) would render negative; since the network model admits no negative injection at a load bus, such values are clipped at zero and the surplus redistributed proportionally. This is a genuine loss of realism — O'ahu does have circuits exporting at midday — and it is visible in the results as a minimum share of exactly 0.00% at WAIANAE. Finally, all bus loads are rescaled so that $\sum_i D_{i,t}$ reproduces the filed net load exactly.

4.4 Verification

Table 2: Verification of the reconstructed field. Only the distribution of demand is altered; its total is preserved.

	v1 (population)	v2 (sector)	v3 (hourly)
Matrix rank	1	1	2
Singular values	—	—	62.7, 5.79, 0.14
Honolulu share, minimum	39.87%	56.89%	52.71%
Honolulu share, maximum	39.87%	56.89%	84.10%
Annual swing (points)	0.00	0.00	31.38
Max error vs filed total (MW)	—	—	3×10^{-4}

The rank is two rather than three because industrial load is modelled flat and therefore contributes no independent temporal variation; the second dimension is entirely residential-versus-commercial. We regard reporting rank two as more honest than engineering a third from data we do not have.

The diurnal pattern is not imposed. Averaged over the year, the Honolulu share runs 55.8% at midnight, 75.4% at noon and 53.7% at 22:00. Two mechanisms push the same way at midday: commercial load peaks in the urban core while distributed PV suppresses net load in the residential periphery.

5 Experimental design

Constraints are added one at a time, so that any movement in a reported statistic is attributable to a single restriction:

- (i) **lp**: $0 \leq P \leq P^{\max}$ — Set 8’s constraint set, as control;
- (ii) **pmin**: binary commitment and (2);
- (iii) **ramp**: (5)–(6);
- (iv) **full**: (4) and start-up costs.

6 Results

6.1 The constraint ladder

Table 3: The constraint ladder under the reconstructed (v3) demand field, full year. ENS is energy not served.

Rung	Hours at a line limit	Hours shedding	ENS (MWh)	Cost (\$M)
lp	94.77%	0.00%	0.0	688.5
pmin	58.93%	0.00%	0.0	693.3
ramp	90.37%	0.61%	33.8	689.0
full	62.66%	0.21%	11.7	689.8

6.2 Ramp, not capacity, is the binding constraint

The system serves all load at the first two rungs. Energy is shed only once (5) is imposed — in 0.61% of hours, about 53 hours per year, totalling 33.8 MWh. Nothing about installed

capacity changes between **pmin** and **ramp**; only the permitted rate of change does. At a 43% reserve margin, the machines exist and cannot arrive in time.

This is the paper’s substantive new result, and it is a category of risk the earlier model could not represent, because a model without temporal adjacency has nothing for a rate limit to bind against. It is also small: 33.8 MWh against annual consumption near 6.5 TWh. It is not zero, and storage is precisely the technology that removes it, which makes storage the natural next study rather than a generic recommendation.

Shedding *falls* from **ramp** to **full**. This is not paradoxical: minimum up-times force units to remain committed through quiet hours, so more capacity is already synchronised and able to move when demand rises. Commitment inertia is a cost in fuel and a benefit in flexibility.

6.3 Congestion: direction without magnitude

The fraction of hours with at least one line at its limit moves substantially across the ladder (94.77% → 58.93% → 90.37% → 62.66%). The pattern is interpretable — committing fewer, larger units concentrates injection and loads the corridors; minimum up-times then spread it again — but by Assumption 2.2 every line rating and reactance producing these numbers is our own. We therefore report the direction and explicitly decline to interpret the magnitude. These are statements about a nine-bus sketch, not about the O’ahu transmission system.

7 Self-audit

Two errors in our own construction were found after the first complete set of results, and both changed the answer. We report them with their effect.

7.1 Reference-calendar misalignment

The OpenEI reference-building simulations run on a calendar year beginning on a Sunday; 2021 begins on a Friday. Both are non-leap, so *dates* correspond exactly and our implementation assumed *weekdays* did too. They are two days apart. The consequence was that simulated Saturday and Sunday were filed as Thursday and Friday: a large office read as busier at the weekend than midweek. The diagnostic signature is unmistakable in the raw profile — the 2 p.m. load for a large office in June reads 1,568, 1,568, 953, 532, 1,566, 1,567, 1,567 kW on days labelled Tuesday through Monday. The weekly pattern is correct; it sits on the wrong days.

The offset was confirmed independently against three building types with a genuine work-week signal (large office, primary school, hospital), all implying +2. Correcting it moved the weekend-to-weekday commercial ratio from 1.14 to 0.63 at the profile level. The hour-of-day curve moved by at most 0.036 of its peak and the peak hour remained 13:00 — so the headline diurnal result was unaffected, while the weekly rhythm had been inverted.

7.2 Headcount as a proxy for energy

The commercial composite was initially weighted by employment alone, which assumes an office worker and a hotel worker represent equal load. EIA CBECS 2018 (table C22 over table B1) gives electricity per worker by principal building activity: 58,888 kW h for food sales, 36,179 for lodging, 20,266 for mercantile, and 6,912 for office — the lowest of any activity measured. Re-weighting by estimated energy moves offices from 39.5% of the commercial blend to 18.4%, with lodging/food and retail each gaining roughly twelve points.

The substantive finding is that O’ahu’s commercial load is *tourism-shaped rather than office-shaped*, and headcount weighting concealed it. The observable consequence is again

the working week: hotels, restaurants and shops trade seven days where offices do not, so the weekend-to-weekday ratio rises from 0.757 to 0.809 in the delivered field. CBECS intensities are national; applied to a tourism economy they likely understate lodging, making this correction conservative.

8 Sensitivity

The allocation of each employment category across building types (Section 4.3) has no source and remains our judgement. Rather than declare it, we tested it: the commercial profile was rebuilt under 300 random within-category mixes and 300 random between-category re-weightings spanning a ninefold range, measuring the maximum deviation of the normalised hour-of-day curve.

Table 4: Sensitivity of the hour-of-day commercial profile. A deviation of 1.000 would indicate complete change of shape.

Perturbation	Max deviation	Peak hour
Equal weights within category	0.007	13:00
Largest building type only	0.055	13:00
Health care $\times 3$	0.070	13:00
Lodging and food $\times 3$	0.090	13:00
Worst of 600 random variants	0.104	13:00

The peak hour never moved. The mechanism is transparent: every commercial building type shares a daytime-peaking profile, and averaging mutually consistent curves cannot produce much disagreement. This contrasts instructively with the calendar error of Section 7, which was *structural* — it relocated load onto the wrong days — and therefore mattered far more than a large error in weights.

The result’s own boundary is equally clear. It holds because nothing in the average runs at night. Uniformed military personnel are excluded from LODES, so Pearl Harbor–Hickam, Schofield Barracks and Kāne‘ohe Marine Corps Base — large, continuously operating loads — contribute nothing to these profiles. That, not the building mix, is the live exposure.

9 Limitations

1. **Operating limits are assumed.** Neither EIA-860 nor EIA-923 carries unit-level minimum stable output or ramp rate, and the IGP workbooks expose no traceable unit table for them. Table 1 therefore rests on technology class conventions. The ramp result of Section 6 depends entirely on these values; obtaining verified operating data is the highest-value improvement available to this model.
2. **The network is assumed** (Assumption 2.2).
3. **Military load is absent**, as above.
4. **Vintage mismatch.** The fleet is 2024 vintage; the load series is 2021. AES Hawaii (180MW of coal) retired in September 2022 and is correctly absent from the fleet, but served the 2021 load. The model therefore asks a post-AES fleet to serve pre-AES demand — a defensible scenario, but not a reconstruction of either year, and it should be named as such.
5. **No forced outages, no $N - 1$, no storage.** Every unit is available in every hour. For an adequacy study this remains the largest single gap.

6. **Reference-building profiles** are typical-year simulations, not O‘ahu’s metered sector loads; industrial load is flat; distributed PV is allocated by population.

10 Conclusion

Modelling O‘ahu’s generators as machines rather than dials does not overturn the Set 8 finding that the assumed demand distribution governs computed risk. It does reveal a failure mode the earlier formulation could not express: at a 43% reserve margin the fleet is unable to follow real demand in roughly 53 hours per year, shedding 33.8 MWh. The constraint that binds is rate, not quantity.

Methodologically, we argue that the rank-one demand field implicit in system-total load data is a more serious limitation than it is usually treated as, and that it can be partially repaired from public data. The reconstruction presented here raises the field to rank two and produces a diurnal rotation of the demand direction of 31.4 percentage points, while preserving the filed system total to 3×10^{-4} MW.

Finally, we note that of the two errors found by self-audit, the damaging one was structural rather than numerical, while a deliberately mis-specified building mix — tested over 600 variants — barely moved the answer. We take this as an argument for auditing the *shape* of an assumption before refining its parameters.

A Reproducibility

All results are reproducible from the project repository. The Set 9 pipeline is:

1. `prepare_demand.py` — static sector reallocation, (8).
2. `prepare_hourly_demand.py` — sector-resolved hourly field, (9). Both scripts download their own sources.
3. `fleet.jl` — case construction; all assumed parameters are declared here and nowhere else.
4. `commitment.jl` — the MILP, (1)–(6).
5. `run_hourly.jl` — the ladder under v1, v2 and v3.

Raw archives are SHA-256 pinned in `SOURCES.md`; the demand reconstruction, its verification and its sensitivity analysis are documented in `DATA_REBUILD.md`.

B Assumed parameters

Table 5: Parameters with no documentary source, collected for challenge.

Parameter	Basis	Value
$P^{\min}/P^{\max}$, oil steam	technology class	0.40
Ramp, oil steam	technology class	30%/h
Minimum up-time, oil steam	convention	8 h
Minimum up-time, municipal waste	continuous refuse feed	24 h
Start-up cost	convention, USD per MW capacity	20–100
Line thermal limit	per 138 kV circuit	200 MW
Line reactance	at 138 kV, 100 MV A base	0.80 Ω /mi
Routing factor	great-circle multiplier	1.25
VOLL	convention	10,000 USD/MWh
Ewa split meridian	by census tract	-158.075°